Mobile Development
Async & Future Builder

LAB 5/W6

3ND YEAR - ENSIA

OBJECTIVES

- Structuring Projects
- Learning how to use Async & Future Builder

TASKS BEFORE THE LAB - WILL BE MARKED.

- You need to create a backend API Endpoint:
  - Feel free to use the code provided from GitHub https://github.com/imed5/mobile_backend_simple_flask
  - Use [Leapcell.io](https://leapcell.io) to make your API endpoint accessible.
- Download the NewsApp with FutureBuilder :
  - Change the code to use your own API Endpoint
  - Execute and run it on your machine or mobile device.

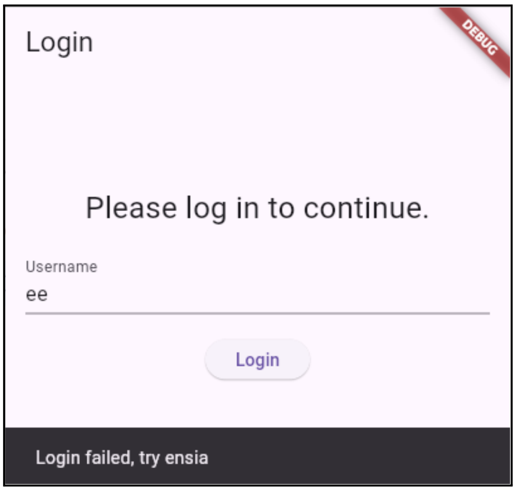
STRICTLY INDIVIDUAL EXERCISES :

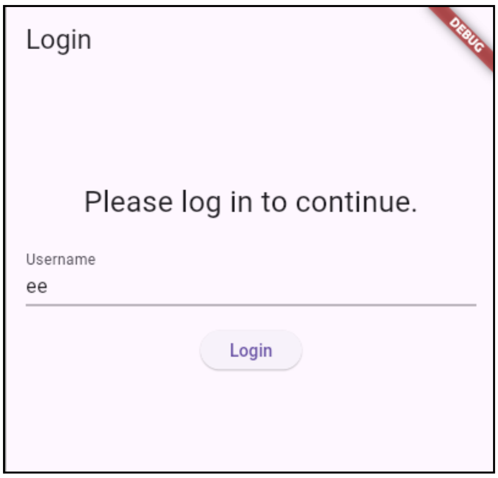
Task / Exercise C1 (Form Validation + Async Coding with Flutter/No Future Builder) [45 mins + 15 mins demonstration by Students] :

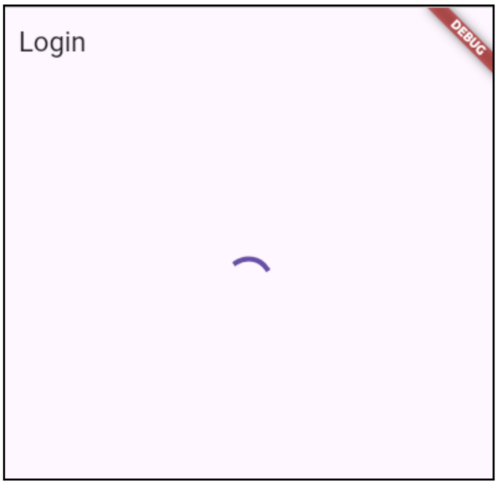
In the same way as the App demonstrated in Class. Develop the following Login App :

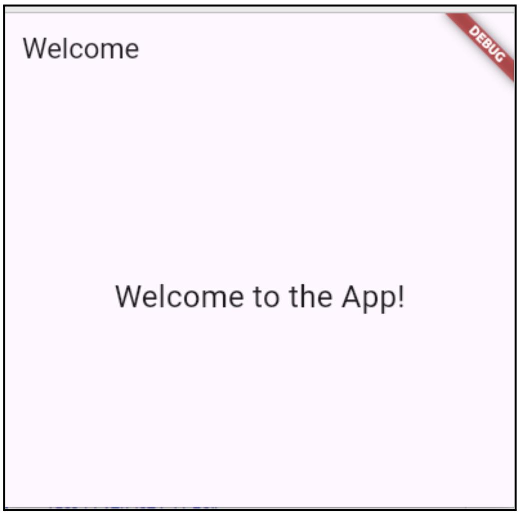
- Simple Login screen, User types the incorrect username (anything not **ensia**), the form is hidden, Circular progress bar is shown for 5 seconds. The form is shown again, with a snackbar message showing the error **as a snackbar**
- If the user enters the username **ensia**, circular progress bar is shown again in place of the form for 5 seconds, afterwards, the user is forwarded to a new screen **WelcomeScreen**
- You need to use form validation to ensure that username is more than 3 letters.
- You need to utilize the following function to fake/simulate the login process.

```
Future<bool> fakeLoginProcess(String username) async {
  await Future.delayed(Duration(seconds: 5));
  return username == 'ensia';
}
```









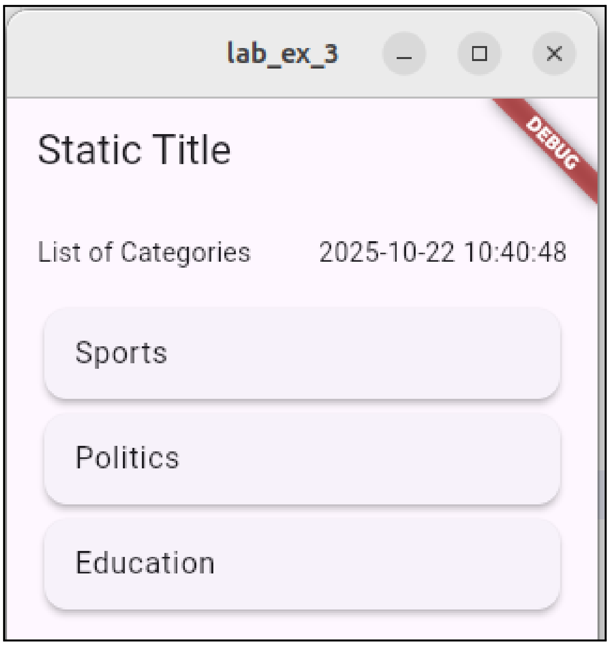
Task / Exercise C2 (Hello World Future Builder) [45 mins + 15 mins demonstration] :

The welcome screen should instead show the list of categories loaded from the following API Endpoint :

https://mobilebackendsimpleflask-imed1024-s73qk5q3.leapcell.dev/news.categories.get

Which returns data shown as below:

```
{
    "title": "List of Categories",
    "time": "2025-10-22 10:40:48",
    "categories": [
        {"id": 1, "name": "Sports"},
        {"id": 2, "name": "Politics"},
        {"id": 3, "name": "Education"}
    ]
}
```



Hint: to speed up your dev, link directly the welcome screen with the Main MaterialApp home property instead of the login screen.

TEAM EXERCISES

Task / Exercise C3 (Progress on MVP) [60 mins]:

Students should continue to work on developing the flutter code/screens for their projects.

Students shouldn't focus on the business logic but just advance with completing the design of all screens as possible.

OPTIONAL EXERCISES

Task / Exercise P1 (Design Patterns)

For the previous exercise C2, use the repo pattern to load the data from the dummy class where data is hardcoded in a simple variable for the case when there is no internet connection. Whilst, if there is an internet connection, use the provided API endpoint to fetch the data.

Task / Exercise P2 (Data Structure : Tetris) [30 mins] (Highly Recommended):

Create a board of 20x30 cells . There is only a single shape of the form **L** which comes from the top center of the board. The shape moves downward with some 500 milliseconds per row.

The user will be able to move the shape left or right **only** within the board (**Not rotation).**

Whenever the shape hits the bottom of the board or another object, it will stop moving. And another **L** shape will drop from the top.

Task / Exercise P3 :

Continue to implement the Tetris game where :

- Able to rotate a shape.
- Whenever a row is full, remove that row, and all objects above that row, will be shifted down.

Task / Exercise P4 :

For the sudoku game, think of the algorithm to generate the board data only.